

Minjae Kim

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SUMMARY

An aspiring pre-med second-year mechanical engineering student with technical and analytical experience across environmental research, biomedical design, and data-driven project work. Skilled in Python, Excel, CAD, and statistical analysis, with a track record of leading multidisciplinary teams and delivering clear, evidence-based insights. Bringing a structured problem-solving approach, strong communication skills, and the ability to turn complex technical findings into practical solutions.

KEY SKILLS

Python

Certified SOLIDWORKS Associate

Matlab

Microsoft Office

Canva

English/Korean Fluency

CPR/AED Certification

National Lifeguard Certification

Swim Instructor Certification

EDUCATION

University of Toronto

April 2025 - April 2029

Bachelor of Applied Science and Engineering in Mechanical Engineering + PEY Co-op

- Intended Minors: Bioengineering and Robotics & Mechatronics
- Relevant Courses: Engineering Analysis (Calculus III), Mechanical Engineering Design (SolidWorks), Probability and Statistics with Engineering Applications, Differential Equations, Material Science

University of Toronto

September 2024 - April 2025

Bachelor of Applied Science and Engineering in Chemical Engineering + PEY Co-op

- Year 1 CGPA: 3.76
- University of Toronto Scholar's Program (\$10K)
- Relevant Courses: Fundamentals of Computer Programming (Python), Engineering Strategies and Practices I & II, Calculus I & II

WORK EXPERIENCE

Miles Nadal Jewish Community Centre

May 2025 - Current

Lifeguard/Swim Instructor

- Supervise aquatic activity of 20+ patrons.
- Performed routine pool maintenance checks and facility cleaning to maintain a 100% health regulation pass rate.

City of Toronto - Environmental Monitoring & Protection

May 2025 - August 2025

Administrative Trainee

- Assist with administrative tasks for the water division using Microsoft Office
- Respond to daily emails
- Assist with a research project for the City's water quality guidelines

PROJECTS

Robot Arm for Cinematography

September 2025 - December 2025

University of Toronto - Mechanical Engineering Design

- Designed and CAD-modelled a 6-DOF cinematography robotic arm in a 4-person team using SolidWorks, integrating revolute joints, gear trains, and motorized actuation to meet smooth-motion and positioning requirements for camera control.
- Developed engineering specifications and performed component selection (materials, motors, gears, joints, and transmission mechanisms) using load analysis and design constraints, resulting in a mechanically feasible and manufacturable robotic arm concept.
- Produced a comprehensive engineering design report outlining requirements, design rationale, calculations, CAD drawings, and trade-off analyses, effectively communicating system architecture and engineering decisions.

Storm Sewer Bylaw Amendment Research

May 2025 - August 2025

City of Toronto - Environmental Monitoring & Protection

- Researched and analyzed the impacts of manganese (Mn), zinc (Zn), copper (Cu), and total suspended solids (TSS) on water quality to assess environmental and regulatory implications.
- Compiled, cleaned, and evaluated 2024 storm and sanitary sewer application data using Excel to identify contaminant trends and assess compliance with existing sewer use bylaws.
- Contributed to the development of a sewer bylaw amendment for the City of Toronto Water Division by providing data-driven findings to support evidence-based policy decisions.

NMES Textile Redesign Project - Team Leader

January 2025 - April 2025

University of Toronto - Engineering Strategies & Practices II

- Led a team of six to develop a 1500-word project requirements document for a biomedical client, ensuring compliance with 5+ industry regulations and stakeholder needs.
- Defined 7 key project objectives, including reducing ICU labour requirements, increasing patient accessibility, and ensuring compliance with ISO and FDA safety standards.
- Coordinated 50+ hours of team research, analysis, and technical writing to establish 6 functional requirements, 7 objectives, and 6 constraints for the device design.
- Delivered and prototyped a working NMES therapy arm brace design that aligns with the client's and stakeholders' requests and constraints.
- Delivered a 10-minute presentation on the final design to an Engineering Manager.

CLUBS

UTBiome

September 2025 - Present

AST Subsystem Member - Darkfield Imaging

- Research dark field microscopy for antimicrobial susceptibility testing applications.
- Analyze different dark field microscopy methods and specifications to recommend a design for the project.
- CAD and built a low budget dark field microscope to capture images for antimicrobial susceptibility testing.

Engineering Athletics Association

May 2025 - Present

Women's Intramural Director

- Coordinate captains and registration for women's intramural sports teams (basketball, soccer, volleyball, field hockey, flag football, lacrosse) in the faculty of engineering.
 - Attend monthly Intramural Student Council meetings and vote on the Intramural handbook, which governs intramural sports at the University of Toronto.
 - Coordinate with other faculties to provide U of T students with recreational sport opportunities.
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